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Food for thought: Physiological considerations for nutritional ergogenic efficacy

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Top-class athletes have optimized their athletic performance largely through adequate training, nutrition, recovery, and sleep. A key component of sports nutrition is the utilization of nutritional ergogenic aids, which may provide a small but significant increase in athletic performance. Over the last decade, there has been an exponential increase in the consumption of nutritional ergogenic aids, where over 80% of young athletes report using at least one nutritional ergogenic aid for training and/or competition. Accordingly, due to their extensive use, there is a growing need for strong scientific investigations validating or invalidating the efficacy of novel nutritional ergogenic aids. Notably, an overview of the physiological considerations that play key roles in determining ergogenic efficacy is currently lacking. Therefore, in this brief review, we discuss important physiological considerations that contribute to ergogenic efficacy for nutritional ergogenic aids that are orally ingested including (1) the impact of first pass metabolism, (2) rises in systemic concentrations, and (3) interactions with the target tissue. In addition, we explore mouth rinsing as an alternate route of ergogenic efficacy that bypasses the physiological hurdles of first pass metabolism via direct stimulation of the central nervous system. Moreover, we provide real-world examples and discuss several practical factors that can alter the efficacy of nutritional ergogenic aids including human variability, dosing protocols, training status, sex differences, and the placebo effect. Taking these physiological considerations into account will strengthen the quality and impact of the literature regarding the efficacy of potential ergogenic aids for top-class athletes.

KEYWORDS
athlete, ergogenic aid, performance, skeletal muscle, sports nutrition, supplementation

1 | INTRODUCTION

Exercise training results in a myriad of adaptations at the cellular and whole-body levels that improve sports performance.[1–3] Top-class athletes also employ several additional modifiable factors including adequate nutrition, recovery practices, and sleep routines to optimize their athletic performance.[4] Athletes, coaches, and support personnel are continually searching for novel strategies that can further improve athletic performance and provide an advantage during competition.[4] For the top-class athlete, marginal performance improvements

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(e.g., 1%–3%) can be the difference between first place and not being on the podium.[5,6] Notably, in elite athletes, a key constituent of sports nutrition is the use of nutritional ergogenic aids to improve an individual's ability to perform an exercise task or recover between training sessions or competitions.[4,7]

Nutritional ergogenic aids can be whole foods, but they are typically specific nutrients or compounds isolated from whole foods and are consumed in quantities that exceed basic nutritional needs.[8,9] Notably, the consumption of supraphysiological amounts of some nutrition-related products has the potential to overcome the body's physiological systems to tightly regulate these compounds in the blood and tissues and alter aspects of metabolism, which may benefit athletic performance. These compounds are legal for consumption by athletes in training and prior to competition and include examples like supplemental creatine, β-alanine, and nitrate.

Athletes have utilized nutritional ergogenic aids for centuries, where ancient records state that athletes and warriors would consume specific parts of animals to gain strength, agility, and speed in preparation for competitions or battles.[10] However, in the past few decades, there has been a dramatic increase in the prevalence of nutritional ergogenic aid use among athletes, where more than 80% of young athletes have reported using at least one ergogenic aid in training and/or competition.[11] Accordingly, due to their extensive use, there is a continual need for strong scientific investigations validating or invalidating the efficacy of novel nutritional ergogenic aids.

There are several excellent publications providing practical guidance for researchers conducting studies designed to examine the efficacy of potential nutritional ergogenic aids for improving athletic performance.[12–15] However, there are also several physiological considerations that can be taken into account when designing studies to examine the ergogenic efficacy of current and novel nutritional ergogenic aids. Therefore, the purpose of this brief review is to provide "food for thought" for researchers performing ergogenic aid research by discussing several key physiological considerations that may contribute to nutritional ergogenic aid efficacy as well as explaining mechanism(s) that may be at play in the laboratory or field. This paper is not meant to extensively examine all potential nutritional ergogenic aids but provides examples of compounds that highlight the physiological considerations that are discussed. Several systematic reviews and meta-analyses exist in the literature to determine if popular nutritional supplements are ergogenic.[16–18]

2 | PHYSIOLOGICAL FACTORS THAT INFLUENCE THE EFFICACY OF ORALLY INGESTED NUTRITIONAL SUPPLEMENTS—THE CLASSIC APPROACH

The purpose of this section is to discuss physiological parameters that influence the potential for ergogenic aid efficacy via the traditional oral route of nutritional ergogenic aid ingestion. Sports nutrition supplements that are consumed orally are subject to first pass metabolism, which plays a critical role in determining the bioavailability profile of these ergogenic aids.[19] Bioavailability is defined as the percentage of the ergogenic aid that reaches the systemic circulation[19] and there are several locations where the orally ingested ergogenic aid could be metabolized or broken down including the mouth, stomach, small intestines, and liver, thereby reducing the bioavailability (Figure 1).

2.1 | Digestive and absorptive systems

First pass metabolism begins in the mouth, which plays key roles in digestion and metabolism.[19,20] Importantly,

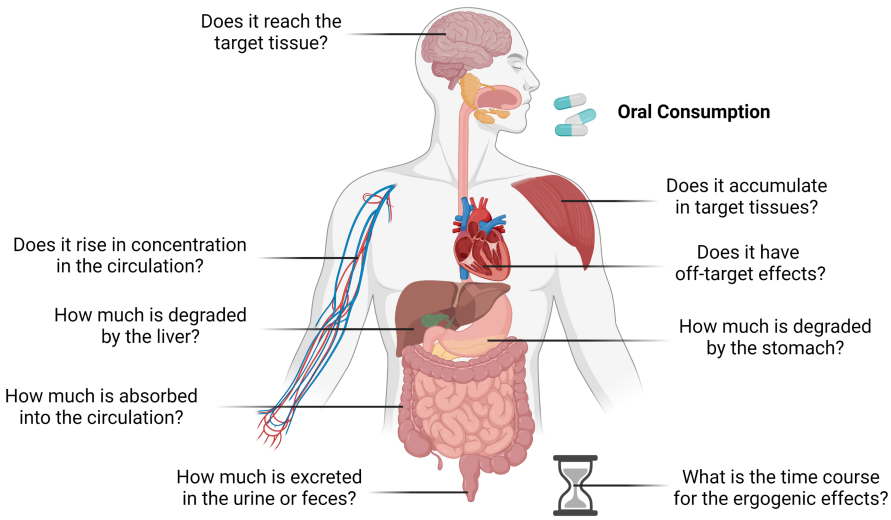

FIGURE 1 Important questions that should be considered regarding bioavailability, rises in systemic concentrations, interactions with target tissues, and ultimately efficacy of orally ingested ergogenic aids. Created with BioRender.com.

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the mouth hosts a plethora of oral bacteria that are critical for the reduction of molecules into other bioactive compounds.[21] Once swallowed, food and food components travel through the esophagus to the stomach. A major challenge for orally administered ergogenic aids is the harsh acidic environment of the stomach (pH ~2), which denatures molecules and significantly reduces their bioavailability.[22] In addition, gastric enzymes (e.g., pepsin, gelatinase) can degrade nutrients,[23,24] such as bioactives found in ergogenic aids. The nutrients that survive the acidic environment of the stomach then enter the small intestines. Though posing a less potent threat than other regions of the digestive system, pancreatic enzymes, such as lipase, trypsin, amylase, and peptidases, are released into the small intestines and influence the breakdown and subsequent bioavailability of nutrients. These enzymes are widely abundant in the duodenum (i.e., the beginning of the small intestines) however, their abundance and activity are significantly reduced in the jejunum and ileum.[25] Moreover, the transit time through the duodenum is short, comprising ~10% of the time spent in the intestinal tract, which minimizes the potential for degradation.[26] However, it is worth considering that the mechanical stress of the intestinal lumen can also degrade nutrients[27] and the mucosal layer lining the gastrointestinal tract can capture and discard foreign molecules, with a particular affinity for hydrophobic molecules.[27]

2.2 | Liver

The liver is considered the predominant site of first pass metabolism.[28] All nutrients absorbed from the gastrointestinal tract enter the liver via the portal vein.[28] In digestion, the liver is a gatekeeper responsible for removing toxins from the body before they enter systemic circulation.[28] Accordingly, the liver degrades numerous compounds into water-soluble by-products, which can be returned to the intestines via bile and excreted. The cytochrome P450 family is the group of enzymes largely responsible for this action in the liver.[28] In addition, the liver plays a critical role in regulating the release of produced (e.g., cholesterol, glucose) and/or stored (e.g., vitamins and minerals, amino acids, glycogen) nutrients for uptake and use by various tissues.[29]

Taken together, the contributions of the digestive and absorptive systems as well as the liver significantly influence the bioavailability of nutrients and bioactives, such as those found in ergogenic aids, and whether these bioactives can elicit a rise in concentration in the systemic circulation.

2.3 | Rise in concentration in the systemic circulation

A rise in concentration in the systemic circulation is an important component of ergogenic aid efficacy for orally ingested supplements, as it is necessary for these ergogenic aids to successfully reach and interact with the target tissue (e.g., skeletal muscle). Determination of plasma concentrations is the gold standard for validating the rise in the concentration of an ergogenic aid, or its breakdown products, in the circulation.[30] Importantly, although compounds can survive first pass metabolism and increase in concentration in the circulation, it is also possible that the compound is rapidly degraded in the circulation or excreted by the kidneys,[31] which could be detected via urinary concentrations (Figure 1). Therefore, it is critical to consider the magnitude of increase in the circulation and the duration that the compound remains elevated in the circulation to elicit a beneficial effect.

2.4 | Delivery and interaction with the target tissue

Finally, efficacious ergogenic aids need to interact with the target tissue, whether this is communication with plasma membrane receptors or uptake of the ergogenic aid into the target tissue. Often, for ergogenic aids that elicit long-term effects, the bioactive ingredient is taken up and stored in the target tissue and impacts cellular metabolism. For example, obtaining muscle biopsies is the gold standard for validating the storage of an ergogenic aid, or its breakdown products, in skeletal muscle.[32] In addition, magnetic resonance spectroscopy has been proven as a valuable noninvasive tool to determine the skeletal muscle content of ergogenic aids.[33,34]

In summary, this section described three physiological considerations that contribute to the efficacy of orally ingested nutritional ergogenic aids: (1) survives first pass metabolism in sufficient quantities, (2) elicits a substantial rise in concentration in the systemic circulation for a reasonable length of time, and (3) interacts with and/or is taken up by the target tissue.

3 | EXAMPLES OF ERGOGENIC AIDS THAT FIT THE CLASSIC APPROACH

This section will discuss examples of orally ingested nutritional ergogenic aids that meet the above considerations,

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are competition legal, and demonstrate a meaningful improvement in athletic performance.

3.1 | Creatine

Creatine plays a critical role in skeletal muscle metabolism as about two-thirds of the creatine in muscle cells is bound to phosphate as phosphocreatine (PCr), a key fuel for rapid energy production.[35,36] Creatine is predominantly stored in skeletal muscle (~95% of total body stores), and accordingly, the main source of dietary creatine is red meat.[37] Interestingly, aerobic and high-intensity interval training (HIIT) do not augment the intramuscular creatine or PCr levels,[1–3] making exogenous creatine supplementation an attractive prospect. Early research demonstrated that consuming an oral 5 g dose of creatine monohydrate resulted in very large elevations in plasma creatine (~20–40 μM to ~600–900 μM), peaking 1 h after ingestion and lasting for 4–5 h.[38] Notably, a 5 g dose of creatine is equivalent to ~1.1 kg of uncooked steak.[38] Therefore, consuming this dose of creatine is a supraphysiological amount, which could not be achieved through a normal diet.

Importantly, in the late 1990s and early 2000s, it was discovered that creatine is transported into skeletal muscle against its concentration gradient by creatine transporters.[37,39] Then, the question became whether exposure of the transporters to these supraphysiological plasma concentrations would result in increases in muscle creatine and PCr contents? It was subsequently shown that a rapid or fast loading regimen where 20 g of creatine was consumed per day for 5 days, in 4 × 5 g doses separated by 4–5 h, resulting in very high plasma creatine concentrations for 16–20 $h \cdot day^{-1}$ and critically, increased muscle creatine content by ~25%–30% and PCr by 10%–15%.[38,40] Most of the increase in skeletal muscle creatine and PCr occurred in the first 2 days, as the majority of the ingested creatine on days 3–5 appeared in the urine.[41] After the initial "fast loading" period, the higher intramuscular creatine levels can be maintained by consuming a single daily dose of ~2 g.[40] Alternately, it has also been demonstrated that "slow loading" by taking a single 3 g dose every day for a period of 28 days can achieve the same final intramuscular accumulations.[40] Critical for athletes, supplemental creatine enhances athletic performance during repeated, short-burst, high-intensity efforts.[42] This is achieved by enhancing the intramuscular PCr stores and by improving the rate of PCr resynthesis in the recovery period between sprint efforts.[43,44] It is also important to note that 10%–15% of people do not respond to supplemental creatine despite large increases in plasma creatine levels, in part due to already high muscle creatine and PCr contents.[44]

3.2 | β-Alanine

After the advent of creatine as an ergogenic aid, researchers continued to explore alternative supplements and pathways that could improve high-intensity exercise performance. During high-intensity exercise, hydrogen ions accumulate, reducing the intramuscular pH, and ultimately contributing to the development of fatigue.[45] Accordingly, carnosine quickly became a target of interest due to its pH buffering capacity in skeletal muscle.[46] Notably, carnosine is more abundant in fast-twitch (type II) muscle fibers, which are predominantly responsible for high-intensity work.[47,48] However, it was soon realized that supplemental carnosine was not the most efficient ergogenic aid, as it was not taken up into skeletal muscle.[48,49] Instead, it was hydrolyzed in the blood to its constituents, β-alanine and histidine, and was expensive to purchase. However, carnosine can be resynthesized in skeletal muscle from β-alanine and histidine via carnosine synthase, where β-alanine is the rate-limiting substrate.[50] Therefore, β-alanine supplementation came into the spotlight as a novel and less-expensive ergogenic aid for high-intensity exercise performance.

The plasma concentration of β-alanine rises in a dose-dependent manner, with peak plasma concentrations occurring ~30 min after ingestion[49,51] and lasting ~2.5 h.[52] β-alanine is administered via a slow-release formula to minimize paresthesia, a known side effect of supplementation.[52] Moreover, if β-alanine is taken chronically at 3.2 $g \cdot day^{-1}$ for 4 weeks, skeletal muscle carnosine concentrations rise, and a greater volume of high-intensity work can be performed.[49] Similar to creatine, after the initial 4-week loading period, the dose can be reduced to 1.6 $g \cdot day^{-1}$, and the intramuscular levels and performance benefits are maintained.[34]

3.3 | Nitrate

Over the last two decades, a growing interest has emerged in the acute and chronic consumption of dietary nitrates, such as beetroot juice, to improve athletic performance.[17] The ergogenic component of dietary nitrate supplementation is attributed to the elevated production of nitric oxide, which is achieved through the systematic reduction of nitrate to nitrite and further to nitric oxide.[17] When dietary nitrates are consumed, they

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successfully survive first pass metabolism, and nitrates are concentrated in the salivary glands, where they are then secreted into the mouth via saliva.[53] Critically, nitrate is reduced to nitrite by salivary bacteria in the mouth. The nitrite is then swallowed and absorbed into the systemic circulation, where it can be broken down to its unstable but biologically active form, nitric oxide, at target tissues.[53]

3.3.1 | Acute supplementation

The plasma levels of nitrate and nitrite are dose-dependent, where higher doses result in higher systemic concentrations.[54] Typically, acute dietary nitrate supplementation is administered in doses of ~8–13 mmol. Pharmacokinetic analyses have demonstrated that nitrate levels reach peak concentrations ~1-h postingestion, and peak nitrite levels occur ~2.5-h postingestion following an acute dose.[54] Dietary nitrate has been shown to improve exercise economy in recreationally trained males by reducing the oxygen cost of submaximal exercise.[17,54–57] This has been proposed to occur through the improvement of either skeletal muscle contractile or mitochondrial efficiency, although the evidence for contractile changes[58–61] is stronger than for mitochondrial efficiency changes.[57,58,62] Importantly, this has been shown to translate to improvements in time trial performance.[55,63,64] Interestingly, the plasma concentrations of nitrate and nitrite have been shown to decrease dramatically following a bout of maximal exercise,[56] suggesting that the nitrate and nitrite are utilized during exercise.

3.3.2 | Chronic supplementation

When consumed chronically (e.g., daily for up to 28 days), nitrate and nitrite levels remain elevated in the systemic circulation.[56] Chronic dietary nitrate supplementation studies have used diverse supplementation protocols, where some studies administer 3–6 mmol daily for 28 days[56] and others administer 26 mmol daily for 14 days.[57,61] Importantly, recent evidence has demonstrated that skeletal muscle houses the necessary machinery for nitrate transport (i.e., sialin) and metabolism (i.e., xanthine oxidoreductase). It was also shown that chronic dietary nitrate supplementation results in the long-term storage of nitrate and nitrite in skeletal muscle.[65,66] Moreover, previous evidence highlights that the oxygen cost of exercise remains reduced with chronic supplementation, and the performance benefits persist.[56,64]

4 | EXAMPLES OF ERGOGENIC AIDS THAT FAIL SOME CONSIDERATIONS

There are examples in the literature where ergogenic aids appear to have the potential to improve performance, but ultimately fail to do so. One example is supplemental taurine, where large increases in plasma concentrations can be produced by ingesting ~1.6 g every 4 h, 2–3 times per day, much like the case for creatine.[67] Taurine is a highly abundant amino acid in skeletal muscle and plays a role in many cellular functions, including calcium handling during contractions.[68] It has been shown that taurine supplementation increases skeletal muscle taurine content in rodents.[69] However, supplementation with 3 doses of 1.6 $g \cdot day^{-1}$ taurine for 7 days in humans failed to alter the skeletal muscle taurine concentration at rest or following exercise, suggesting that the transporter involved in taurine movement into the cell was unaffected by the large plasma concentrations for several hours a day.[70] Unsurprisingly, cycling performance in well-trained triathletes and cyclists was not improved with taurine supplementation.[67] However, it remains possible that elevated taurine levels in the blood could positively affect other organs in the body (e.g., brain, autoimmune system) to improve athletic performance and this possibility has been reviewed recently.[71]

A second example is supplemental carnitine, where the difficulty resides in increasing the concentration of free carnitine in the blood, despite serious loading attempts. Carnitine plays a critical role in transferring fat from the cytoplasm to the mitochondria and in buffering the rise in acetyl-CoA at the onset of exercise in skeletal muscle.[72] An early study demonstrated modest increases of ~30% in the blood of healthy young males following a supplemental regimen, where 6 $g \cdot day^{-1}$ carnitine was consumed for 14 days. However, no increases in skeletal muscle carnitine or whole-body metabolism occurred.[73] More recent efforts have reported ~20% increases in muscle following ~6 months of supplementing with 2 g of carnitine along with ~80 g of carbohydrates twice a day in healthy young males.[74] However, the work in this area has demonstrated the difficulty in increasing plasma carnitine levels and altering the skeletal muscle carnitine content and is not a practical paradigm for athletes to follow.

These are examples where a supplement managed to escape first pass metabolism in sufficient quantities to elicit a potential ergogenic effect but ultimately failed to significantly affect the target organ or improve performance during exercise.

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5 | EXAMPLES OF ERGOGENIC AIDS THAT FAIL ALL CONSIDERATIONS

There are many examples of potential ergogenic aids that do not survive first pass metabolism and do not elicit increases in concentration in the blood. One example is supplementing with oral pyruvate, as early studies from one laboratory suggested a potential performance enhancement.[75] However, pyruvate ingested acutely at tolerable levels up to 25 g did not elicit a rise in concentration in the blood.[76] Moreover, there was no improvement in high-intensity cycling performance to exhaustion in well-trained cyclists who ingested 7 g pyruvate for 7 days.[76] Interestingly, it has been proposed that oral pyruvate may undergo decarboxylation in the stomach and intestines,[76] as it is not excreted in the urine[76] or the feces.[75] Mechanistically, it is not clear how the ingestion of a 3-carbon molecule like pyruvate that does not get into the circulation would be preferable to 5- or 6-carbon sources of carbohydrates, which readily enter the circulation.

A second example is a compound called microhydrin, which is purported to contain silica hydride bonds (Si-H) that would buffer hydrogen ions as they are released into the blood during high-intensity exercise and could potentially increase time trial performance.[77] However, direct infrared spectrometry analysis showed a complete absence of Si-H bonds in microhydrin, when it was dissolved in water as suggested by the manufacturer.[78] Additionally, time trial performance and whole-body metabolism were unaffected following the oral ingestion of 9.6 g of microhydrin administered over 48 h compared with a placebo in well-trained cyclists.[78]

Taken together, when evaluating the efficacy of novel orally ingested ergogenic aids, it is important to experimentally consider these physiological checkpoints (Figure 2). Does the supplement have an ergogenic effect? Does the bioactive nutrient enter and rise in concentration in the systemic circulation? Does the bioactive nutrient interact with the target tissue (e.g., skeletal muscle)? What are the mechanisms of action?

6 | ACTIVATION OF ORAL RECEPTORS LINKED TO THE CENTRAL NERVOUS SYSTEM

Traditionally, nutritional ergogenic aids have been orally ingested and are subject to first pass metabolism. Recently, a growing body of research has investigated the efficacy of nutritional ergogenic aids that bypass first pass metabolism and the associated physiological barriers to ergogenic efficacy via alternate routes. Mouth rinsing with compounds such as caffeine,[79,80] carbohydrates,[81–83]

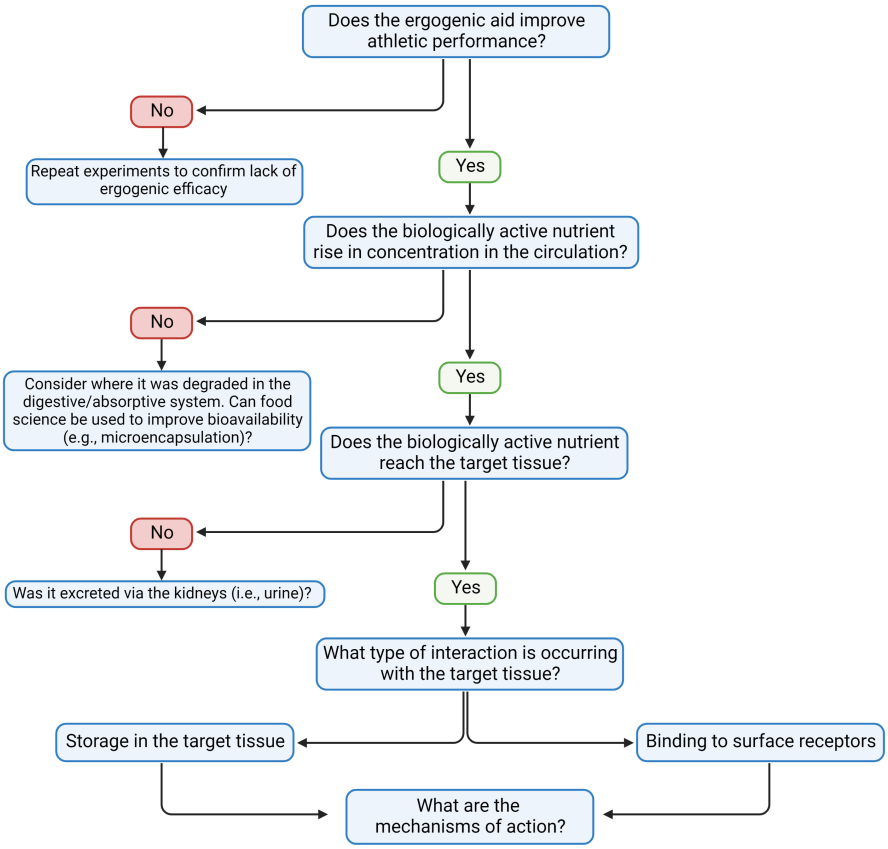

FIGURE 2 Decision flow chart for considering the efficacy of ergogenic aids in laboratory and/or field research. Created with BioRender.com.

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or menthol[84–86] are relatively novel approaches to improve athletic performance and have recently garnered considerable interest within the athletic community. Mouth rinsing involves swishing a concentrated solution around the oral cavity for a period of ~5–20 s and subsequently spitting out the fluid (i.e., none of the fluid is swallowed). This method has been shown to stimulate sensors in the oral cavity to communicate with different regions of the central nervous system, particularly the brain, to elicit changes in behavior and ultimately improve athletic performance[79,81,84] (Figure 1). Specifically, carbohydrate, caffeine, and menthol mouth rinsing have been shown to stimulate reward centers in the brain,[80,87,88] which can improve motivation, reduce the perception of effort, and ultimately enhance athletic performance.[79,81,84]

6.1 | Carbohydrate mouth rinsing

The first mouth rinsing experiment was performed in 2004 by Carter et al.[89] using carbohydrates, where they showed an improvement in 1 h cycling time trial performance. This study transformed the ergogenic field and sparked a new area of nutritional ergogenic aid research. A few years later, a crucial follow-up study by Chambers et al.[87] provided the first mechanistic insights underpinning the ergogenic effects of carbohydrate mouth rinsing, where they showed stimulation of sweet taste receptors in the oral cavity, which activated various reward centers in the brain.[90] Importantly, in the same timeframe, it was also confirmed that carbohydrate mouth rinsing did not alter plasma glucose concentrations, suggesting the ergogenic effect must be mediated by the central nervous system.[91] To date, most studies exploring the effect of carbohydrate mouth rinsing on athletic performance have produced consistent findings, supporting a positive ergogenic effect.[81–83,89]

6.2 | Caffeine mouth rinsing

The intriguing impact of carbohydrate mouth rinsing on athletic performance led to increased interest in alternative formulations, including caffeine mouth rinsing. Caffeine has been shown to interact with bitter taste receptors in the oral cavity to stimulate reward centers in the brain.[92] Although caffeine mouth rinsing appears to improve cognitive performance,[80,93,94] the impact on athletic performance is less convincing, as some studies suggest an improvement[79,80] and others show no effect.[95,96] However, caffeine mouth rinsing may hold value for performance in team sports that may involve a higher cognitive demand (e.g., decision-making, reaction time).

6.3 | Menthol mouth rinsing

Recently, researchers have been interested in the effect of menthol mouth rinsing on athletic performance. Menthol interacts with cold temperature sensors in the oral cavity, which can affect an individual's alertness and motivation.[88] Moreover, the interaction between menthol and cold temperature sensors in the mouth can also play a role in thermoregulatory control. Specifically, menthol mouth rinsing can improve ratings of thermal comfort in hot environments, and there is mounting evidence that performance-enhancing effects can be achieved.[97–99]

Taken together, recent advances have highlighted that nutritional ergogenic aids can elicit beneficial effects on athletic performance without oral ingestion. These novel routes of administration can bypass the physiological hurdles of first pass metabolism associated with orally ingested ergogenic aids.

7 | PRACTICAL PHYSIOLOGICAL CONSIDERATIONS THAT ALTER ERGOGENIC EFFICACY

The study of nutritional ergogenic aids and athletic performance is a fruitful area of research. However, research areas that receive a lot of attention often feature a vast array of strong and weak study designs. This section provides some physiological considerations and practical advice for researchers aiming to improve the validity and strength of their ergogenic research.

7.1 | Human variability

As with most aspects of human physiology and metabolism, there is high interindividual variability in the responses to supplementation with ergogenic aids.[38,100–102] This can include differences in first pass metabolism, rises in concentration in the circulation, interaction with target tissues, storage, and ultimately the overall achieved performance benefit. For example, a few studies have reported large interindividual variability in the intramuscular accumulation of creatine following supplementation (e.g., 110–160 mmol·kg dry mass^{-1}).[38,101] Interestingly, but perhaps unsurprisingly, there is a positive relationship between the amount of creatine and PCr accumulated in skeletal muscle and the performance benefit attained.[101] Therefore, supplementation with ergogenic aids, in this case creatine, produces benefits that are greater in some individuals compared with others as well as the potential for both responders and non-responders. This consideration is crucial for researchers, as it highlights the

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importance of utilizing the correct sample size to detect statistically significant changes in investigations with ergogenic aids. It also brings to light the importance of transparent methodologies and repeating study designs to determine whether a scientific finding is reproducible. For example, the first study investigating supplementation with exogenous ketones demonstrated an improvement in athletic performance in elite athletes.[103] However, subsequent studies have been unable to replicate these findings.[104–106]

7.2 | Dosing protocol

Interindividual variability in ergogenic studies may be partially attributed to the dosing protocol, where a multitude of factors can be modified to alter the efficacy of the nutritional ergogenic aids.

7.2.1 | Dose, timing, and frequency

How much should be taken? When is the best time to take it? How often should it be taken? When setting out to validate a nutritional ergogenic aid, pharmacokinetic studies utilizing serial blood sampling coupled with exercise performance assessments will provide valuable insights into these fundamental questions. The pharmacokinetic profile will show if increasing doses result in increased blood concentrations or if it is a saturable process. In addition, the pharmacokinetic profile gives insights into the time course for rises and falls in plasma concentrations. For acute supplementation protocols, the rise in the pharmacokinetic profile will indicate the best time to take the supplement, which coincides with peak plasma concentrations around the onset of exercise testing, training, or competition. For chronic supplementation protocols, the fall in the pharmacokinetic profile indicates an ideal time to provide an additional dose to maintain high systemic concentrations of the bioactive ingredient.

7.2.2 | Absolute versus relative dosing

Another important consideration is that many ergogenic aids are dosed in absolute quantities (e.g., mg) due to practical usefulness, however, this will significantly alter the relative dose (e.g., $mg \cdot kg^{-1}$) depending on the individual. For example, if a 200 mg dose of caffeine is administered to a 60 kg male and an 85 kg male, the relative doses are 3.3 and 2.4 $mg \cdot kg^{-1}$, respectively. Therefore, when performing ergogenic research, it is important to consider, in the context of the research question, whether it is better to choose the practical simplicity of an absolute dose versus maintaining the same relative dose across the cohort.

7.2.3 | Nutrient co-ingestion

Is there an optimal dietary strategy to maximize the supplement benefits? For example, with creatine supplementation, the optimal intake strategy is co-ingestion with carbohydrates.[41] This has been attributed to the rise in insulin with carbohydrate co-ingestion, which stimulates the uptake of creatine into skeletal muscle.[41,107] However, the amount of carbohydrates that must be ingested may make this practice untenable. For researchers, it may be beneficial to consider these interactions, as it provides an opportunity to standardize the co-ingestion of these critical supporting nutrients as well as afford the greatest opportunity to observe an ergogenic effect of supplementation.

Conversely, are there any compounds that can blunt or inhibit the beneficial effects of the ergogenic aid? Previous research has shown that the utilization of antibacterial mouthwashes blunts the ergogenic effects of dietary nitrate supplementation in a strength-dependent manner.[108,109] Antibacterial mouthwashes destroy the oral bacteria necessary for the conversion of nitrate to nitrite in the oral cavity, where stronger mouthwashes elicit greater impairments.[108,109] Accordingly, when mouthwash is used around the time of supplementation with dietary nitrate, the rise in the concentration of nitrite in the circulation is minimized, which may reduce the ergogenic potential of dietary nitrate supplementation.[108] Accordingly, researchers need to consider the positive and negative interactions with other components of the diet and/or lifestyle of the participants, and standardization protocols should be put in place to accommodate these potential interactions.

7.3 | Training status

An important question is whether the ergogenic aid is more effective for recreationally active individuals or elite athletes. First, it is generally more difficult to improve the athletic performance of an already elite individual. However, in these cohorts, as little as a 1% improvement could be the difference between a podium finish or not, and this may be difficult to detect statistically.[5,6] A critical example of this stems from dietary nitrate research, where a growing body of evidence suggests that the ergogenic benefits are predominantly isolated to recreationally active individuals[54,55,57] and there may be minimal to no beneficial effect in elite athletes.[110–113]

However, this is not always the case, as some ergogenic aids favor adaptations in trained individuals. For example,

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it appears that accrual of muscle carnosine with β-alanine supplementation is better achieved in trained compared to untrained muscles.[114] Taken together, it is important to consider the target population of the research question and to acknowledge that there may be divergent benefits of the ergogenic aid depending on the training status of the participants.

7.4 | Sport-specific ergogenic effects

Does the ergogenic effect produce beneficial effects for endurance events, sprint events, and/or stop-and-go sports? By having a clear rationale and potential mechanism of action for the ergogenic aid under investigation, it is easier to select an appropriate exercise modality. However, it is important to be flexible in physiological reasoning. For example, dietary nitrate supplementation was originally proposed to be beneficial for endurance exercise performance, as nitric oxide availability could potentially improve blood flow and oxygen delivery to exercising skeletal muscle.[115] However, recent evidence suggests that in highly trained populations, the ergogenic effects of dietary nitrate are not observed in elite endurance athletes,[110,111] but only in stop-and-go team sport athletes.[116,117] Notably, the ergogenic effects of dietary nitrate supplementation may primarily occur in fast-twitch (type IIb) muscle fibers,[118] potentially explaining the observed benefits in stop-and-go team sport athletes versus endurance athletes who have mainly type I and type IIa fibers. In addition, elite endurance athletes may have elevated circulating nitric oxide levels, which could minimize the benefit of exogenous nitrate supplementation compared to stop-and-go team sport athletes.[119]

7.5 | Sex differences

Another important question is whether men and women respond differently to these supplements. Evidence suggests that men and women obtain similar benefits from creatine[120–122] and β-alanine[122–124] supplementation. However, there is emerging evidence that recreationally active and trained women do not experience the ergogenic effects of dietary nitrate supplementation ,[61,119,125–127] despite a higher relative dose, whereas the ergogenic benefits are consistently reported in recreationally active men.[54,55,57] This highlights the complexity of sex-specific responses in human physiology and metabolism and supports the ongoing calls for greater inclusion of female participants in sports science and ergogenic aid research.[128,129]

7.6 | Alternate measurements

Confirming a physiological response to ergogenic aid supplementation, such as plasma analysis or interaction with the central nervous system, can often be costly, invasive, and inaccessible. If an ergogenic aid, at a verified dose, has been repeatedly validated to elicit a rise in concentration in the circulation or elicit ergogenic effects on the central nervous system, it is possible to explore alternate methods to confirm a physiological effect. For example, dietary nitrate supplementation has been repeatedly shown to increase plasma concentrations of nitrate and nitrite[54,56] and to elicit the physiological response of blood pressure lowering.[130,131] Accordingly, it is feasible to use the non-invasive and cost-effective measurement of blood pressure lowering over a 2–2.5 h window to demonstrate the efficacy of the supplementation program.[132] Second, for ergogenic aids that stimulate the central nervous system, validation of the mechanism of action requires advanced techniques, such as functional magnetic resonance imaging[87] or electroencephalography.[80] Accordingly, if the ergogenic aid has previously been verified to positively impact the central nervous system through activation of motivation and reward centers in the brain using advanced methodologies, a cost-effective and non-invasive solution may be to acquire ratings of perceived exertion throughout the exercise task of interest.

Taken together, in scenarios where blood sampling and advanced analyses are not feasible, researchers are encouraged to seek out creative and valid alternatives to showcase the efficacy of ergogenic aids in experimental settings.

7.7 | Placebo and blinding effects

Sports performance is complex and multifaceted. An emerging area in sports science is the quantitative exploration of the placebo effect as it relates to athletic performance. A meta-analysis of studies systematically testing the placebo effect found that most placebo interventions produced a 1%–5% improvement in athletic performance in sub-elite athletes,[133] highlighting the power of psychology in sport. This is a critical consideration for researchers conducting ergogenic research, as psychology can play a role in controlled laboratory trials. For example, the phrasing that researchers use when communicating with a participant can significantly influence the results of the study.[134] Consider the following scenarios: (1) after consuming a caffeine supplement, athletes run faster if they think they consumed caffeine compared to when they are falsely told it is a placebo,[135] (2) athletes perform better if they receive a supplement and are told

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it will enhance performance versus when they are told it will have no effect,[136,137] and (3) athletes perform better when they consume a placebo and are told it will enhance performance compared to when they are told it will impair performance.[138] Notably, these placebo effect studies were performed under conditions where the participants were under the impression that they were receiving a known efficacious ergogenic aid (e.g., caffeine or sodium bicarbonate). Interestingly, the placebo effect is not present in situations where the potential ergogenic effect does not work and/or the potential ergogenic benefits are not widely known by the participants.[67] In the same vein, unsuccessful blinding can significantly bias the outcomes of experimental trials,[139] where participants may perform better when they believe they have received a treatment that is known to elicit ergogenic effects. Together, the athlete and/or participant's beliefs can strongly influence their performance, and care should be taken to standardize communication with participants between trials and conditions[134] as well as to optimize the treatment blinding.[139]

8 | CONCLUSIONS

In summary, an orally ingested putative nutritional ergogenic aid would hopefully meet the following physiological considerations: (1) survive first pass metabolism in sufficient quantities, (2) elicit a substantial rise in concentration in the systemic circulation for a reasonable length of time, and (3) interact with and/or is taken up by the target tissue. In these scenarios, the nutritional ergogenic aid has the potential to overcome the body's physiological systems to tightly regulate these compounds in the blood and tissues, which sets up the possibility of providing a meaningful performance-enhancing effect. However, there are also novel nutritional ergogenic aids that can directly stimulate the central nervous system to elicit a performance-enhancing effect, and ultimately bypass some of the physiological hurdles associated with first pass metabolism. Taking these physiological considerations into account when conducting research studies will strengthen the quality and impact of the literature regarding the efficacy of potential ergogenic aids for top-class athletes.

9 | PERSPECTIVE

Nutritional ergogenic aid research is highly impactful for, and readily accessed by, the general population (e.g., coaches, athletes, weekend warriors, fitness enthusiasts, etc.). Therefore, it is important to encourage and support researchers in the field to strive to produce studies with high impact, validity, and strength to provide the general population with strong physiological evidence supporting or refuting the efficacy of nutritional ergogenic aids. This brief review may help researchers consider and tackle the physiological hurdles that arise in nutritional ergogenic aid research by providing "food for thought" regarding the physiological factors influencing the efficacy of nutritional ergogenic aids.

AUTHOR CONTRIBUTIONS

KAW and LLS designed and drafted the manuscript. Both authors approved the final version of the manuscript.

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CONFLICT OF INTEREST

The authors have no conflicts of interest to declare.

DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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REFERENCES

1. Chesley A, Heigenhauser GJ, Spriet LL. Regulation of muscle glycogen phosphorylase activity following short-term endurance training. *Am J Physiol*. 1996;270:E328-E335.
2. Talanian JL, Galloway SDR, Heigenhauser GJF, Bonen A, Spriet LL. Two weeks of high-intensity aerobic interval training increases the capacity for fat oxidation during exercise in women. *J Appl Physiol*. 2007;102:1439-1447.
3. Perry CGR, Heigenhauser GJF, Bonen A, Spriet LL. High-intensity aerobic interval training increases fat and carbohydrate metabolic capacities in human skeletal muscle. *Appl Physiol Nutr Metab*. 2008;33:1112-1123.
4. Maughan RJ, Burke LM, Dvorak J, et al. IOC consensus statement: dietary supplements and the high-performance athlete. *Int J Sport Nutr Exerc Metab*. 2018;28:104-125.
5. Hopkins WG, Hawley JA, Burke LM. Design and analysis of research on sport performance enhancement. *Med Sci Sports Exerc*. 1999;31:472-485.

16000838, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.14307, Wiley Online Library on [23/03/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

6. Malcata RM, Hopkins WG. Variability of competitive performance of elite athletes: a systematic review. *Sports Med*. 2014;44:1763-1774.
7. Kreider RB, Wilborn CD, Taylor L, et al. ISSN exercise & sport nutrition review: research & recommendations. *J Int Soc Sports Nutr*. 2010;7:7.
8. Williams MH, Leutholtz BC. Nutritional ergogenic aids. In: Maughan RJ, ed. *Nutrition in Sport*. John Wiley & Sons, Ltd; 2000:357-366.
9. Close GL, Kasper AM, Walsh NP, Maughan RJ. 'Food first but not always food only': recommendations for using dietary supplements in sport. *Int J Sport Nutr Exerc Metab*. 2022;32:371-386.
10. Applegate EA, Grivetti LE. Search for the competitive edge: a history of dietary fads and supplements. *J Nutr*. 1997;127:869S-873S.
11. Jovanov P, Đorđić V, Obradović B, et al. Prevalence, knowledge and attitudes towards using sports supplements among young athletes. *J Int Soc Sports Nutr*. 2019;16:27.
12. Burke LM. Practical issues in evidence-based use of performance supplements: supplement interactions, repeated use and individual responses. *Sports Med*. 2017;47:79-100.
13. Burke LM, Peeling P. Methodologies for investigating performance changes with supplement use. *Int J Sport Nutr Exerc Metab*. 2018;28:159-169.
14. Betts JA, Gonzalez JT, Burke LM, et al. PRESENT 2020: text expanding on the checklist for proper reporting of evidence in sport and exercise nutrition trials. *Int J Sport Nutr Exerc Metab*. 2020;30:2-13.
15. Close GL, Kasper AM, Morton JP. From paper to podium: quantifying the translational potential of performance nutrition research. *Sports Med*. 2019;49:25-37.
16. Bonilla DA, Kreider RB, Stout JR, et al. Metabolic basis of creatine in health and disease: a bioinformatics-assisted review. *Nutrients*. 2021;13:1238.
17. Jones AM, Thompson C, Wylie LJ, Vanhatalo A. Dietary nitrate and physical performance. *Annu Rev Nutr*. 2018;38:303-328.
18. Matthews JJ, Artioli GG, Turner MD, Sale C. The physiological roles of carnosine and β-alanine in exercising human skeletal muscle. *Med Sci Sports Exerc*. 2019;51:2098-2108.
19. Herman TF, Santos C. First pass effect. In: *StatPearls*. StatPearls Publishing; 2022.
20. Patricia JJ, Dhamoon AS. Physiology, digestion. In: *StatPearls*. StatPearls Publishing; 2022.
21. Carpenter GH. Salivary factors that maintain the normal oral commensal microflora. *J Dent Res*. 2020;99:644-649.
22. Moroz E, Matoori S, Leroux JC. Oral delivery of macromolecular drugs: where we are after almost 100 years of attempts. *Adv Drug Deliv Rev*. 2016;101:108-121.
23. Homayun B, Lin X, Choi HJ. Challenges and recent progress in oral drug delivery systems for biopharmaceuticals. *Pharmaceutics*. 2019;11:129.
24. Kong F, Singh RP. Disintegration of solid foods in human stomach. *J Food Sci*. 2008;73:R67-R80.
25. Layer P, Go VL, DiMagno EP. Fate of pancreatic enzymes during small intestinal aboral transit in humans. *Am J Physiol*. 1986;251:G475-G480.
26. Fallingborg J, Christensen LA, Ingeman-Nielsen M, Jacobsen BA, Abildgaard K, Rasmussen HH. pH-profile and regional transit times of the normal gut measured by a radiotelemetry device. *Aliment Pharmacol Ther*. 1989;3:605-613.
27. Ensign LM, Cone R, Hanes J. Oral drug delivery with polymeric nanoparticles: the gastrointestinal mucus barriers. *Adv Drug Deliv Rev*. 2012;64:557-570.
28. Vaja R, Rana M. Drugs and the liver. *Anaesth Intensive Care Med*. 2020;21:517-523.
29. Kalra A, Yetiskul E, Wehrle CJ, Tuma F. Physiology, liver. In: *StatPearls*. StatPearls Publishing; 2022.
30. Thiha A, Ibrahim F. A colorimetric enzyme-linked immunosorbent assay (ELISA) detection platform for a point-of-care dengue detection system on a lab-on-compact-disc. *Sensors (Basel)*. 2015;15:11431-11441.
31. Masereeuw R, Russel FG. Mechanisms and clinical implications of renal drug excretion. *Drug Metab Rev*. 2001;33:299-351.
32. Greene J, Louis J, Korostynska O, Mason A. State-of-the-art methods for skeletal muscle glycogen analysis in athletes—the need for novel non-invasive techniques. *Biosensors (Basel)*. 2017;7:11.
33. Vandenberghe K, Goris M, Van Hecke P, Van Leemputte M, Vangerven L, Hespel P. Long-term creatine intake is beneficial to muscle performance during resistance training. *J Appl Physiol*. 1997;83:2055-2063.
34. Stellingwerff T, Anwander H, Egger A, et al. Effect of two β-alanine dosing protocols on muscle carnosine synthesis and washout. *Amino Acids*. 2012;42:2461-2472.
35. Spriet L. Anaerobic metabolism during exercise. In: McConell G, ed. *Exercise Metabolism*. Springer; 2022:51-70.
36. Hargreaves M, Spriet LL. Skeletal muscle energy metabolism during exercise. *Nat Metab*. 2020;2:817-828.
37. Schoch RD, Willoughby D, Greenwood M. The regulation and expression of the creatine transporter: a brief review of creatine supplementation in humans and animals. *J Int Soc Sports Nutr*. 2006;3:60-66.
38. Harris RC, Söderlund K, Hultman E. Elevation of creatine in resting and exercised muscle of normal subjects by creatine supplementation. *Clin Sci (Lond)*. 1992;83:367-374.
39. Guerrero-Ontiveros ML, Wallimann T. Creatine supplementation in health and disease. Effects of chronic creatine ingestion in vivo: down-regulation of the expression of creatine transporter isoforms in skeletal muscle. *Mol Cell Biochem*. 1998;184:427-437.
40. Hultman E, Söderlund K, Timmons JA, Cederblad G, Greenhaff PL. Muscle creatine loading in men. *J Appl Physiol*. 1996;81:232-237.
41. Green AL, Hultman E, Macdonald IA, Sewell DA, Greenhaff PL. Carbohydrate ingestion augments skeletal muscle creatine accumulation during creatine supplementation in humans. *Am J Physiol*. 1996;271:E821-E826.
42. Casey A, Constantin-Teodosiu D, Howell S, Hultman E, Greenhaff PL. Creatine ingestion favorably affects performance and muscle metabolism during maximal exercise in humans. *Am J Physiol*. 1996;271:E31-E37.
43. Yquel RJ, Arsac LM, Thiaudière E, Canioni P, Manier G. Effect of creatine supplementation on phosphocreatine resynthesis, inorganic phosphate accumulation and pH during intermittent maximal exercise. *J Sports Sci*. 2002;20:427-437.
44. Greenhaff PL, Bodin K, Soderlund K, Hultman E. Effect of oral creatine supplementation on skeletal muscle phosphocreatine resynthesis. *Am J Physiol*. 1994;266:E725-E730.

16000838, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.14307, Wiley Online Library on [23/03/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

45. Metzger JM, Moss RL. Effects of tension and stiffness due to reduced pH in mammalian fast- and slow-twitch skinned skeletal muscle fibres. *J Physiol*. 1990;428:737-750.
46. Culbertson JY, Kreider RB, Greenwood M, Cooke M. Effects of Beta-alanine on muscle carnosine and exercise performance:a review of the current literature. *Nutrients*. 2010;2:75-98.
47. Harris RC, Wise JA, Price KA, Kim HJ, Kim CK, Sale C. Determinants of muscle carnosine content. *Amino Acids*. 2012;43:5-12.
48. Bauer K, Schulz M. Biosynthesis of carnosine and related peptides by skeletal muscle cells in primary culture. *Eur J Biochem*. 1994;219:43-47.
49. Hill CA, Harris RC, Kim HJ, et al. Influence of beta-alanine supplementation on skeletal muscle carnosine concentrations and high intensity cycling capacity. *Amino Acids*. 2007;32:225-233.
50. Artioli GG, Gualano B, Smith A, Stout J, Lancha AHJ. Role of β-alanine supplementation on muscle carnosine and exercise performance. *Med Sci Sports Exerc*. 2010;42:1162-1173.
51. Harris RC, Tallon MJ, Dunnett M, et al. The absorption of orally supplied β-alanine and its effect on muscle carnosine synthesis in human vastus lateralis. *Amino Acids*. 2006;30:279-289.
52. Décombaz J, Beaumont M, Vuichoud J, Bouisset F, Stellingwerff T. Effect of slow-release β-alanine tablets on absorption kinetics and paresthesia. *Amino Acids*. 2012;43:67-76.
53. Webb AJ, Patel N, Loukogeorgakis S, et al. Acute blood pressure lowering, vasoprotective and anti-platelet properties of dietary nitrate via bioconversion to nitrite. *Hypertension*. 2008;51:784-790.
54. Wylie LJ, Kelly J, Bailey SJ, et al. Beetroot juice and exercise: pharmacodynamic and dose-response relationships. *J Appl Physiol*. 2013;115:325-336.
55. Lansley KE, Winyard PG, Bailey SJ, et al. Acute dietary nitrate supplementation improves cycling time trial performance. *Med Sci Sports Exerc*. 2011;43:1125-1131.
56. Wylie LJ, Ortiz de Zevallos J, Isidore T, et al. Dose-dependent effects of dietary nitrate on the oxygen cost of moderate-intensity exercise: acute vs. chronic supplementation. *Nitric Oxide*. 2016;57:30-39.
57. Whitfield J, Ludzki A, Heigenhauser GJF, et al. Beetroot juice supplementation reduces whole body oxygen consumption but does not improve indices of mitochondrial efficiency in human skeletal muscle. *J Physiol*. 2016;594:421-435.
58. Bailey SJ, Fulford J, Vanhatalo A, et al. Dietary nitrate supplementation enhances muscle contractile efficiency during knee-extensor exercise in humans. *J Appl Physiol*. 2010;109:135-148.
59. Kumar R, Coggan AR, Ferreira LF. Nitric oxide and skeletal muscle contractile function. *Nitric Oxide*. 2022;122-123:54-61.
60. Whitfield J, Gamu D, Heigenhauser GJF, et al. Beetroot juice increases human muscle force without changing Ca2+-handling proteins. *Med Sci Sports Exerc*. 2017;49:2016-2024.
61. Wickham KA, McCarthy DG, Pereira JM, et al. No effect of beetroot juice supplementation on exercise economy and performance in recreationally active females despite increased torque production. *Physiol Rep*. 2019;7:e13982.
62. Ntessalen M, Procter NEK, Schwarz K, et al. Inorganic nitrate and nitrite supplementation fails to improve skeletal muscle mitochondrial efficiency in mice and humans. *Am J Clin Nutr*. 2020;111:79-89.
63. Cermak NM, Gibala MJ, Van LLJC. Nitrate supplementation's improvement of 10-km time-trial performance in trained cyclists. *Int J Sport Nutr Exerc Metab*. 2012;22:64-71.
64. Wong TH, Sim A, Burns SF. The effects of nitrate ingestion on high-intensity endurance time-trial performance: a systematic review and meta-analysis. *J Exerc Sci Fit*. 2022;20:305-316.
65. Wylie LJ, Park JW, Vanhatalo A, et al. Human skeletal muscle nitrate store: influence of dietary nitrate supplementation and exercise. *J Physiol*. 2019;597:5565-5576.
66. Nyakayiru JM, Jonvik KL, Pinckaers PJM, Senden J, Van LLJC, Verdijk LB. No effect of acute and 6-day nitrate supplementation on VO2 and time-trial performance in highly trained cyclists. *Int J Sport Nutr Exerc Metab*. 2017;27:11-17.
67. Rutherford JA, Spriet LL, Stellingwerff T. The effect of acute taurine ingestion on endurance performance and metabolism in well-trained cyclists. *Int J Sport Nutr Exerc Metab*. 2010;20:322-329.
68. Huxtable RJ. Physiological actions of taurine. *Physiol Rev*. 1992;72:101-163.
69. Yatabe Y, Miyakawa S, Miyazaki T, Matsuzaki Y, Ochiai N. Effects of taurine administration in rat skeletal muscles on exercise. *J Orthop Sci*. 2003;8:415-419.
70. Galloway SDR, Talanian JL, Shoveller AK, Heigenhauser GJF, Spriet LL. Seven days of oral taurine supplementation does not increase muscle taurine content or alter substrate metabolism during prolonged exercise in humans. *J Appl Physiol*. 2008;105:643-651.
71. Kurtz JA, VanDusseldorp TA, Doyle JA, Otis JS. Taurine in sports and exercise. *J Int Soc Sports Nutr*. 2021;18:39.
72. Howlett RA, Heigenhauser GJ, Hultman E, Hollidge-Horvat MG, Spriet LL. Effects of dichloroacetate infusion on human skeletal muscle metabolism at the onset of exercise. *Am J Physiol*. 1999;277:E18-E25.
73. Barnett C, Costill DL, Vukovich MD, et al. Effect of L-carnitine supplementation on muscle and blood carnitine content and lactate accumulation during high-intensity sprint cycling. *Int J Sport Nutr*. 1994;4:280-288.
74. Wall BT, Stephens FB, Constantin-Teodosiu D, Marimuthu K, Macdonald IA, Greenhaff PL. Chronic oral ingestion of l-carnitine and carbohydrate increases muscle carnitine content and alters muscle fuel metabolism during exercise in humans. *J Physiol*. 2011;589:963-973.
75. Stanko RT, Robertson RJ, Spina RJ, Reilly JJ, Greenawalt KD, Goss FL. Enhancement of arm exercise endurance capacity with dihydroxyacetone and pyruvate. *J Appl Physiol*. 1990;68:119-124.
76. Morrison MA, Spriet LL, Dyck DJ. Pyruvate ingestion for 7 days does not improve aerobic performance in well-trained individuals. *J Appl Physiol*. 2000;89:549-556.
77. Purdy Lloyd KL, Wasmund W, Smith L, Raven PB. Clinical effects of a dietary antioxidant silicate supplement, microhydrin((R)), on cardiovascular responses to exercise. *J Med Food*. 2001;4:151-159.
78. Glazier LR, Stellingwerff T, Spriet LL. Effects of microhydrin supplementation on endurance performance and metabolism in well-trained cyclists. *Int J Sport Nutr Exerc Metab*. 2004;14:560-573.
79. da Silva WF, Lopes-Silva JP, Camati Felippe LJ, Ferreira GA, Lima- Silva AE, Silva-Cavalcante MD. Is caffeine mouth rinsing an effective strategy to improve physical and cognitive

16000838, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.14307, Wiley Online Library on [23/03/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

performance? A systematic review. *Crit Rev Food Sci Nutr*. 2021;63:438-446.
80. De Pauw K, Roelands B, Knaepen K, Polfliet M, Stiens J, Meeusen R. Effects of caffeine and maltodextrin mouth rinsing on P300, brain imaging, and cognitive performance. *J Appl Physiol*. 2015;118:776-782.
81. de Ataide e Silva T, Di Cavalcanti Alves de Souza ME, de Amorim JF, Stathis CG, Leandro CG, Lima-Silva AE. Can carbohydrate mouth rinse improve performance during exercise? A systematic review. *Nutrients*. 2013;6:1-10.
82. Rollo I, Williams C, Gant N, Nute M. The influence of carbohydrate mouth rinse on self-selected speeds during a 30-min treadmill run. *Int J Sport Nutr Exerc Metab*. 2008;18:585-600.
83. Devenney S, Collins K, Shortall M. Effects of various concentrations of carbohydrate mouth rinse on cycling performance in a fed state. *Eur J Sport Sci*. 2016;16:1073-1078.
84. Gavel EH, Hawke KV, Bentley DJ, Logan-Sprenger HM. Menthol mouth rinsing is more than just a mouth wash—swilling of menthol to improve physiological performance. *Front Nutr*. 2021;8:691695.
85. Crosby S, Butcher A, McDonald K, Berger N, Bekker PJ, Best R. Menthol mouth rinsing maintains relative power production during three-minute maximal cycling performance in the heat compared to cold water and placebo rinsing. *Int J Environ Res Public Health*. 2022;19:3527.
86. Stevens CJ, Best R. Menthol: a fresh ergogenic aid for athletic performance. *Sports Med*. 2017;47:1035-1042.
87. Chambers ES, Bridge MW, Jones DA. Carbohydrate sensing in the human mouth: effects on exercise performance and brain activity. *J Physiol*. 2009;587:1779-1794.
88. Eccles R. Role of cold receptors and menthol in thirst, the drive to breathe and arousal. *Appetite*. 2000;34:29-35.
89. Carter JM, Jeukendrup AE, Jones DA. The effect of carbohydrate mouth rinse on 1-h cycle time trial performance. *Med Sci Sports Exerc*. 2004;36:2107-2111.
90. Nelson G, Hoon MA, Chandrashekar J, Zhang Y, Ryba NJ, Zuker CS. Mammalian sweet taste receptors. *Cell*. 2001;106:381-390.
91. Pottier A, Bouckaert J, Gilis W, Roels T, Derave W. Mouth rinse but not ingestion of a carbohydrate solution improves 1-h cycle time trial performance. *Scand J Med Sci Sports*. 2010;20:105-111.
92. Chandrashekar J, Mueller KL, Hoon MA, et al. T2Rs function as bitter taste receptors. *Cell*. 2000;100:703-711.
93. Pomportes L, Brisswalter J, Casini L, Hays A, Davranche K. Cognitive performance enhancement induced by caffeine, carbohydrate and guarana mouth rinsing during submaximal exercise. *Nutrients*. 2017;9:589.
94. Van Cutsem J, De Pauw K, Marcora S, Meeusen R, Roelands B. A caffeine-maltodextrin mouth rinse counters mental fatigue. *Psychopharmacology (Berl)*. 2018;235:947-958.
95. Marinho AH, Mendes EV, Vilela RA, Bastos-Silva VJ, Araujo GG, Balikian P. Caffeine mouth rinse has no effects on anaerobic energy yield during a Wingate test. *J Sports Med Phys Fitness*. 2020;60:69-74.
96. Figueiredo N, Queiroz M, Felício FP, et al. Acute caffeine mouth rinsing does not improve 10-km running performance in CYP1A2 C-allele carriers. *Clin Nutr ESPEN*. 2021;42:93-97.
97. Jeffries O, Goldsmith M, Waldron M. L-Menthol mouth rinse or ice slurry ingestion during the latter stages of exercise in the heat provide a novel stimulus to enhance performance despite elevation in mean body temperature. *Eur J Appl Physiol*. 2018;118:2435-2442.
98. Flood TR, Waldron M, Jeffries O. Oral l-menthol reduces thermal sensation, increases work-rate and extends time to exhaustion, in the heat at a fixed rating of perceived exertion. *Eur J Appl Physiol*. 2017;117:1501-1512.
99. Mündel T, Jones DA. The effects of swilling an L(−)-menthol solution during exercise in the heat. *Eur J Appl Physiol*. 2010;109:59-65.
100. Pickering C, Kiely J. Are the current guidelines on caffeine use in sport optimal for everyone? Inter-individual variation in caffeine ergogenicity, and a move towards personalised sports nutrition. *Sports Med*. 2018;48:7-16.
101. Casey A, Greenhaff PL. Does dietary creatine supplementation play a role in skeletal muscle metabolism and performance? *Am J Clin Nutr*. 2000;72:607S-617S.
102. Barwood MJ, Gibson OR, Gillis DJ, et al. Menthol as an ergogenic aid for the Tokyo 2021 Olympic games: an expert-led consensus statement using the modified Delphi method. *Sports Med*. 2020;50:1709-1727.
103. Cox PJ, Kirk T, Ashmore T, et al. Nutritional ketosis alters fuel preference and thereby endurance performance in athletes. *Cell Metab*. 2016;24:256-268.
104. Poffé C, Ramaekers M, Bogaerts S, Hespel P. Exogenous ketosis impacts neither performance nor muscle glycogen breakdown in prolonged endurance exercise. *J Appl Physiol*. 2020;128:1643-1653.
105. Evans M, Egan B. Intermittent running and cognitive performance after ketone ester ingestion. *Med Sci Sports Exerc*. 2018;50:2330-2338.
106. Whitfield J, Lm B, Aka M, et al. Acute ketogenic diet and ketone ester supplementation impairs race walk performance. *Med Sci Sports Exerc*. 2021;53:776-784.
107. Pittas G, Hazell MD, Simpson EJ, Greenhaff PL. Optimization of insulin-mediated creatine retention during creatine feeding in humans. *J Sports Sci*. 2010;28:67-74.
108. McDonagh S, Wylie L, Winyard P, Vanhatalo A, Jones A. The effects of chronic nitrate supplementation and the use of strong and weak antibacterial agents on plasma nitrite concentration and exercise blood pressure. *Int J Sports Med*. 2015;36:1177-1185.
109. Woessner M, Smoliga JM, Tarzia B, Stabler T, Van Bruggen M, Allen JD. A stepwise reduction in plasma and salivary nitrite with increasing strengths of mouthwash following a dietary nitrate load. *Nitric Oxide*. 2016;54:1-7.
110. Boorsma RK, Whitfield J, Spriet LL. Beetroot juice supplementation does not improve performance of elite 1500-m runners. *Med Sci Sports Exerc*. 2014;46:2326-2334.
111. Peacock O, Tjønna AE, James P, et al. Dietary nitrate does not enhance running performance in elite cross-country skiers. *Med Sci Sports Exerc*. 2012;44:2213-2219.
112. Burke LM, Hall R, Heikura IA, et al. Neither beetroot juice supplementation nor increased carbohydrate oxidation enhance economy of prolonged exercise in elite race walkers. *Nutrients*. 2021;13:2767.
113. Porcelli S, Ramaglia M, Bellistri G, et al. Aerobic fitness affects the exercise performance responses to nitrate supplementation. *Med Sci Sports Exerc*. 2015;47:1643-1651.
114. Bex T, Chung W, Baguet A, et al. Muscle carnosine loading by beta-alanine supplementation is more pronounced in trained vs. untrained muscles. *J Appl Physiol*. 2014;116:204-209.

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115. Bailey SJ, Winyard P, Vanhatalo A, et al. Dietary nitrate supplementation reduces the O2 cost of low-intensity exercise and enhances tolerance to high-intensity exercise in humans. *J Appl Physiol*. 2009;107:1144-1155.
116. Wylie LJ, Mohr M, Krustrup P, et al. Dietary nitrate supplementation improves team sport-specific intense intermittent exercise performance. *Eur J Appl Physiol*. 2013;113:1673-1684.
117. Macuh M, Knap B. Effects of nitrate supplementation on exercise performance in humans: a narrative review. *Nutrients*. 2021;13:3183.
118. Jones AM, Ferguson SK, Bailey SJ, Vanhatalo A, Poole DC. Fiber type-specific effects of dietary nitrate. *Exerc Sport Sci Rev*. 2016;44:53-60.
119. Wickham KA, Spriet LL. No longer beeting around the bush: a review of potential sex differences with dietary nitrate supplementation. *Appl Physiol Nutr Metab*. 2019;44:1-10.
120. Eckerson JM. Creatine as an ergogenic aid for female athletes. *Strength Cond J*. 2016;38:14-23.
121. Candow DG, Forbes SC, Roberts MD, et al. Creatine o'clock: does timing of ingestion really influence muscle mass and performance? *Front Sports Act Living*. 2022;4:4.
122. Murphy MJ, Rushing BR, Sumner SJ, Hackney AC. Dietary supplements for athletic performance in women: beta-alanine, caffeine, and nitrate. *Int J Sport Nutr Exerc Metab*. 2022;32:311-323.
123. Stout JR, Cramer JT, Zoeller RF, et al. Effects of beta-alanine supplementation on the onset of neuromuscular fatigue and ventilatory threshold in women. *Amino Acids*. 2007;32:381-386.
124. Kresta JY, Oliver JM, Jagim AR, et al. Effects of 28 days of beta-alanine and creatine supplementation on muscle carnosine, body composition and exercise performance in recreationally active females. *J Int Soc Sports Nutr*. 2014;11:55.
125. Forbes SPA, Spriet LL. Potential effect of beetroot juice supplementation on exercise economy in well-trained females. *Appl Physiol Nutr Metab*. 2022;47:106-109.
126. Buck CL, Henry T, Guelfi K, Dawson B, McNaughton LR, Wallman K. Effects of sodium phosphate and beetroot juice supplementation on repeated-sprint ability in females. *Eur J Appl Physiol*. 2015;115:2205-2213.
127. Glaister M, Pattison JR, Muniz-Pumares D, Patterson SD, Foley P. Effects of dietary nitrate, caffeine, and their combination on 20-km cycling time trial performance. *J Strength Cond Res*. 2015;29:165-174.
128. Costello JT, Bieuzen F, Bleakley CM. Where are all the female participants in sports and exercise medicine research? *Eur J Sport Sci*. 2014;14:847-851.
129. Smith ES, McKay AKA, Kuikman M, et al. Auditing the representation of female versus male athletes in sports science and sports medicine research: evidence-based performance supplements. *Nutrients*. 2022;14:953.
130. Hobbs DA, George TW, Lovegrove JA. The effects of dietary nitrate on blood pressure and endothelial function: a review of human intervention studies. *Nutr Res Rev*. 2013;26:210-222.
131. Ashworth A, Bescos R. Dietary nitrate and blood pressure: evolution of a new nutrient? *Nutr Res Rev*. 2017;30:208-219.
132. Wickham KA, Steele SW, Cheung SS. Effects of acute dietary nitrate supplementation on cold-induced vasodilation in healthy males. *Eur J Appl Physiol*. 2021;121:1431-1439.
133. Beedie CJ, Foad AJ. The placebo effect in sports performance. *Sports Med*. 2009;39:313-329.
134. Roelands B, Hurst P. The placebo effect in sport: how practitioners can inject words to improve performance. *Int J Sports Physiol Perform*. 2020;15:765-766.
135. Hurst P, Schipof-Godart L, Hettinga F, Roelands B, Beedie C. Improved 1000-m running performance and pacing strategy with caffeine and placebo: a balanced placebo design study. *Int J Sports Physiol Perform*. 2019;15:483-488.
136. McClung M, Collins D. 'Because I know it will!': placebo effects of an ergogenic aid on athletic performance. *J Sport Exerc Psychol*. 2007;29:382-394.
137. Saunders B, de Oliveira LF, da Silva RP, et al. Placebo in sports nutrition: a proof-of-principle study involving caffeine supplementation. *Scand J Med Sci Sports*. 2017;27:1240-1247.
138. Hurst P, Foad A, Coleman D, Beedie C. Athletes intending to use sports supplements are more likely to respond to a placebo. *Med Sci Sports Exerc*. 2017;49:1877-1883.
139. Hróbjartsson A, Emanuelsson F, Skou Thomsen AS, Hilden J, Brorson S. Bias due to lack of patient blinding in clinical trials. A systematic review of trials randomizing patients to blind and nonblind sub-studies. *Int J Epidemiol*. 2014;43:1272-1283.

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